



Title: Process Guideline - Thermoplastic Injection Molding	
DOCUMENT NUMBER: 930-00164	REVISION: 01
PROCESS OWNER: ADVANCED MANUFACTURING ENGINEERING	

1. INTRODUCTION	2
2. PROCESS OVERVIEW	2
2.1. INJECTION MOLDING EQUIPMENT	3
2.2. INJECTION MOLDS	3
2.3. TYPES OF GATES	9
2.4. TYPES OF INJECTION MOLDING PROCESS	15
3. SURFACE FINISHES	18
3.1. SANDING AND POLISHING	18
3.2. EDM SPARK EROSION	19
3.3. MEDIA BLASTING	19
3.4. CHEMICAL PHOTOETCHING	20
3.5. LASER ETCHING	20
4. PART DESIGN GUIDELINES	21
4.1. PARTING LINE	21
4.2. UNDERCUTS	24
4.3. DRAFT ANGLE	24
4.4. RIBS, CORRUGATIONS AND GUSSETS	26
4.5. WALL THICKNESS	30
4.6. RADII AND CHAMFERS	31
4.7. BOSSES	33
4.8. DEFAULT SPECIFICATIONS FOR PLASTIC TOLERANCES	35

1. INTRODUCTION

Injection Molding is the most common method for mass manufacturing of plastic products. Examples include automotive parts, packaging, toys, cases for electronics and a large variety of other parts.

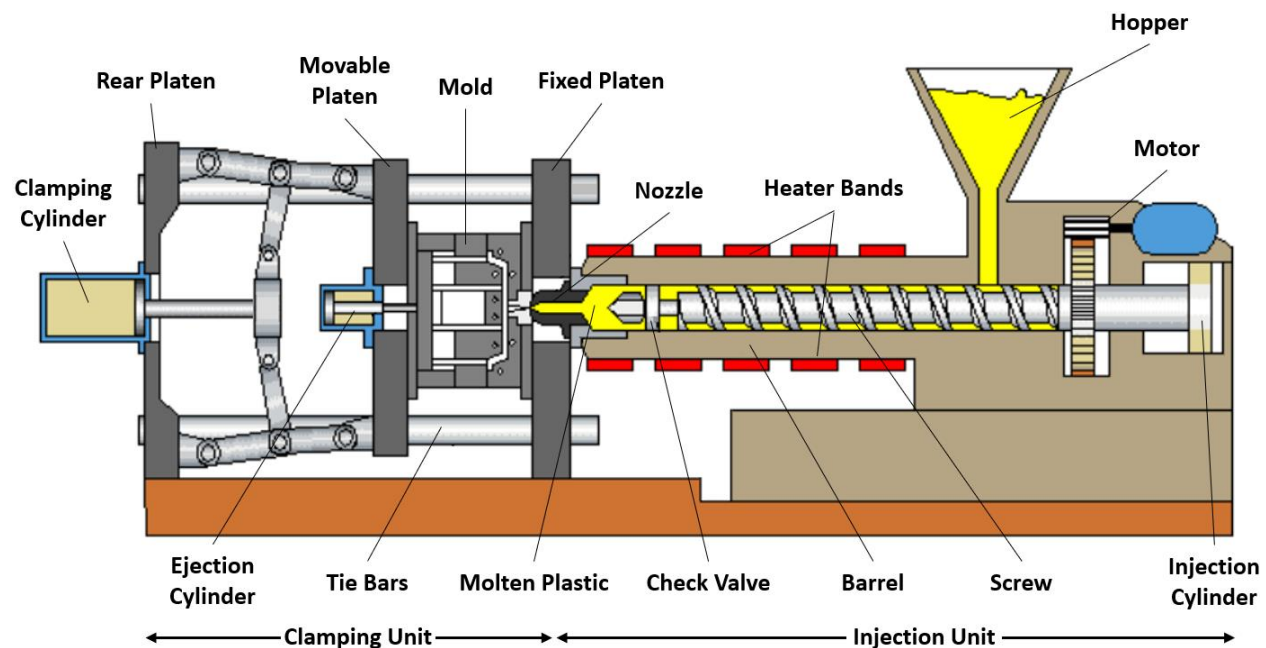
This document will describe the different types of injection molding process, explore types of molds and their parts, explore surface finishes and provide guidelines for part design.

This document will not provide guidelines for tooling design, in regards to the cooling, ejection and venting systems or hydraulic or pneumatic mechanisms. This document will not cover complicated part design such as auto screwing molds, elbow molds or slides on lifter mechanisms.

2. PROCESS OVERVIEW

The injection molding process basically consists of compressing molten plastic into a mold, let it cool and remove it from the mold.

An injection molding machine, usually known as injection press, is divided in two fundamental units: the injection unit and the clamping unit. In the injection unit the solid plastic pellets are fed into the barrel of the injection unit through a hopper. Inside the barrel, a screw transports the pellets forward. Heaters wrapped around the barrel heat up the plastic material and as the pellets are moved forward by the screw, more heat is generated by the friction between the plastic and the screw and barrel causing the pellets to melt. Once enough molten plastic is in front of the screw it rams forward like a plunger of a syringe. In a matter of seconds, the screw injects the molten plastic from the injection unit into the clamping unit, inside an empty part of the mold, called the cavity. The plastic solidifies, usually in under a minute, the mold opens and the part is ejected. The mold then closes and the process repeats again.



2.1. INJECTION MOLDING EQUIPMENT

Hopper: It holds the material that will be processed

Barrel: It has a cylindrical shape and contains the screw that will plasticize the material

Screw: It conveys, plasticizes and is used to inject the molten material into the mold cavity through the nozzle

Heater Bands: It heats the barrel

Injection Cylinder: It pushes the screw in order to inject the molten material into the mold cavity

Nozzle: It is the channel through which the molten material is injected from the barrel, by the screw, into the mold cavity. Usually it contains a needle to avoid the material to return to the barrel

Check Valve: It prevents the molten material to return to the screw during the injection process

Fixed Platen: It is used to clamp the cavity side of the mold

Movable Platen: It is used to clamp the core side of the mold

Rear Platen: It is a fixed platen that supports the tie bars and clamping cylinder

Clamping Cylinder: It is the cylinder that will open and close the mold as well provide the clamping force to keep the mold closed during the injection process. It is also common to have a toggle mechanism to apply the clamping force

Ejection Cylinder: It drives the mechanisms that will eject the part from the mold

Tie Bars: It supports and keep the platens aligned

2.2. INJECTION MOLDS

An injection mold is a tool used to produce injection molded parts. It consists of a series of parts that allow the molten plastic material to be formed and cooled to create a desired component.

2.2.1. Mold Material

Injection molds are usually made of steel or aluminum. The choice of which material is used to build an injection mold depends on several factors, such as:

- **Tooling cost** - Advantage: It depends on which grade of steel or aluminum is chosen. Usually aluminum is cheaper than steel used for injection molds, but there are grades of steel that can be at the same price range of aluminum. One example is free machining steel, that is cheap and easy to machine. The overall value and return on investment of an aluminum mold and a steel mold can vary greatly, based on the usage of the mold (is it going to be a very high volume - millions of parts per year or is it going to be a low volume production rate - thousand or parts per year?), the intended production lifespan, material to be molded, etc.

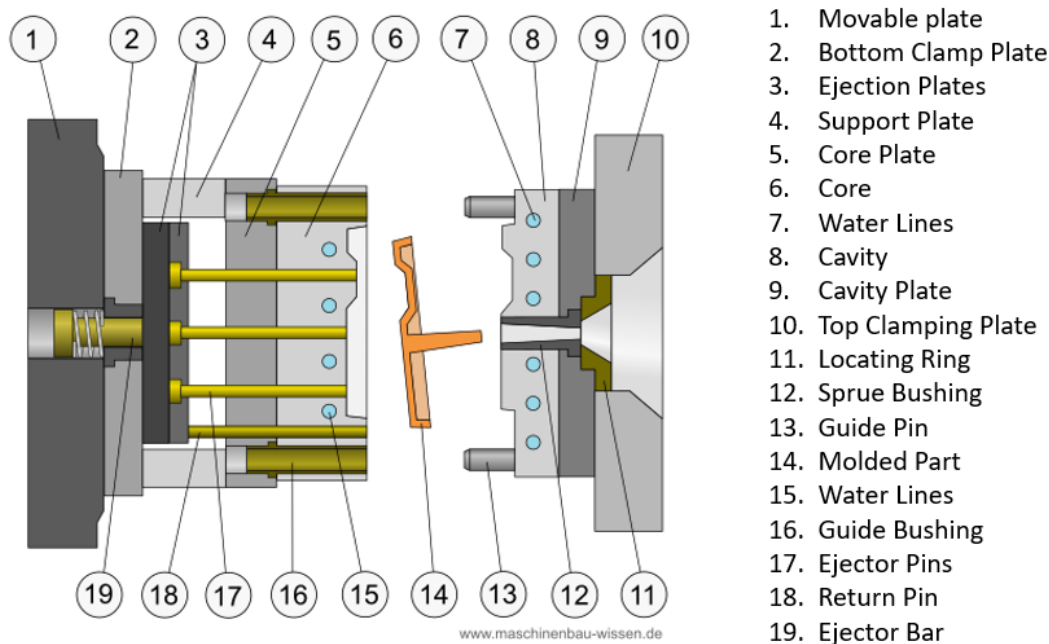
- **Cost per part** - Advantage: Toss-up. This factor depends on your intended production run, and the overall lifespan that you hope to get out of your part. For shorter runs, the lower upfront costs of aluminum molds can lead to lower overall cost per part. If you intend to use your mold for high volumes in the millions and tens of millions of parts, you'll see the investment in a steel mold pay off. What's more, the cost per part will decrease as you get a much longer service life. It's critically important to plan how you'll use your mold in advance, to get the greatest advantage in this area.
- **Suitability for low volumes** - Advantage: Aluminum. As mentioned above, aluminum molds will almost always offer better value for low-volume production runs — due to their lower upfront costs and ability to reliably produce parts into the thousands and tens of thousands of pieces. As always, review the specifics of your project to ensure that, for instance, aluminum is suitable for the material you're using.
- **Suitability for high volumes** - Advantage: Steel. For molds intended for high volume and multiple production runs, a steel construction will typically be your better bet. Your higher upfront investment is rewarded with a piece of equipment that can reliably last for years with proper maintenance. The strength of steel is unparalleled for use in these types of longer production runs.
- **Heating and cooling times** - Advantage: Aluminum. With a much higher rate of heat dissipation than steel, aluminum molds can heat and cool much more quickly than steel molds — typically up to seven times. Cooling time, especially, makes up a significant portion of overall cycle time in injection molding. Thus, choosing an aluminum mold when appropriate can offer major benefits in cycle time which, in turn, means that you're able to produce more parts more quickly.
- **Suitability for advanced resins** - Advantage: Steel. While both steel and aluminum are usually suitable for a broad range of standard injection molding resins, steel can offer an advantage if you're working with more complex or advanced formulations — such as those reinforced with glass, fiber or other additives. A relatively softer metal like aluminum runs a greater risk of scratches or other damage from certain types of additives, which can affect the finish and texture of the final part.
- **Shrink, warp and other defects** - Advantage: Aluminum. The superior heat dissipation of aluminum means that the mold is better able to approach uniform heating and cooling times — and to do so more quickly — which provides an advantage in decreasing the number of defective and rejected parts. Non-uniform heating and cooling are among the biggest factors in defects such as sink marks, voids and burn marks. When used in the right application, aluminum molds can provide even better cost advantages due to lower rejection rates for parts.
- **Ease of modification and repair** - Advantage: Aluminum. Damaged or deformed steel molds can be difficult and costly to repair, depending on the hardness of the material. Sometimes, a new mold may be required in such cases. Aluminum molds are usually more receptive to repair and, as a softer material, can be more easily modified in cases where production errors may have occurred or design changes are required. One way to make future modifications easier is to use inserts for areas of high probability of changes, no matter what material is used to build the mold. In this case only the inserts will be replaced, making the whole modification process simpler.
- **Suitability for detailed features** - Advantage: Steel. When extremely fine, detailed features are required — such as those that push the boundaries of injection mold design best practices, e.g., thin, non-uniform walls, less rounded angles and tighter, narrower areas of the mold cavity —

steel molds will often provide better results than aluminum. The strength and hardness of the steel means that it's better able to hold a shape for those extremely high-precision areas.

- **Durability** - Advantage: Steel. The key benefit of steel molds is their vastly superior durability, over any other material available. When long production runs are expected, and the mold is intended for repeated use over many years, steel is the clear choice. The higher upfront investment required more than pays itself back over the repeated use and tens of millions of parts that the mold is capable of producing. Steel can also lower parts per cost far more than any other material allows. The flip side of this advantage is the risk of using an over-engineered solution for your application. Essentially, if you don't require those long production runs and repeated uses, steel is less likely to prove a worthy investment.

2.2.2. Parts of an injection mold

This section describes the main components of an injection mold.

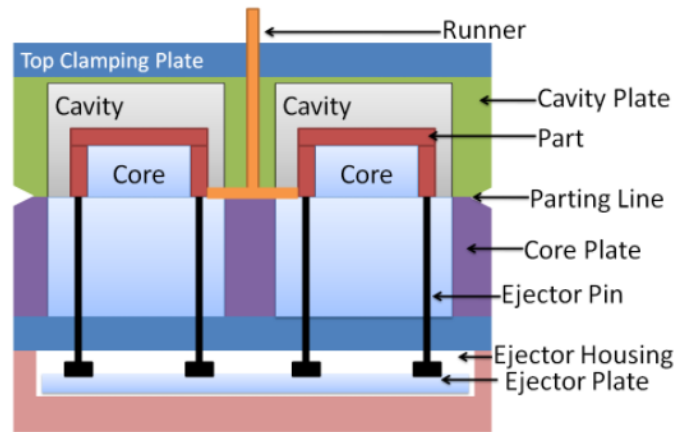


2.2.3. Types of Molds

The selection of the type of injection mold depends on the part geometry, part design requirements, tool design, production volumes and budget. The type of injection mold affects component quality and manufacturing cost. This section describes the various types of injection molds and their components.

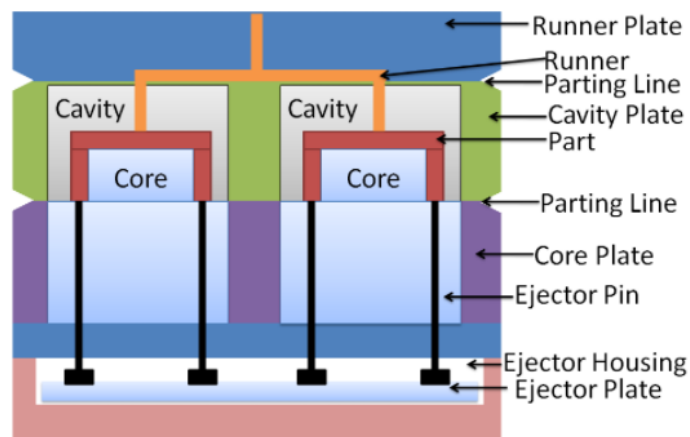
- **Mold Opening Mechanism**
 - **Two Plate Mold**

Two plate molds are the most commonly used type of injection mold. They consist of one parting plane where the mold splits. In a multiple cavity tool, runner and gate must be in the parting plane to ensure runner and gate ejection when mold opens.



- **Three Plate Mold**

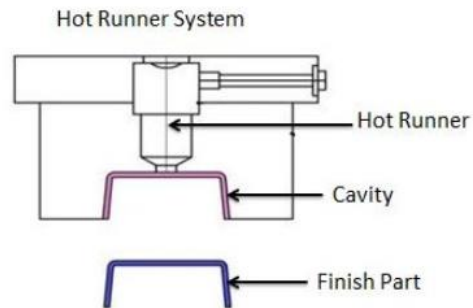
Three plate molds consist of two parting planes and mold splits in three sections. Runner can be located on different parting lines, so they can be ejected separately, turning possible automatic degating.



- **Feeding System**

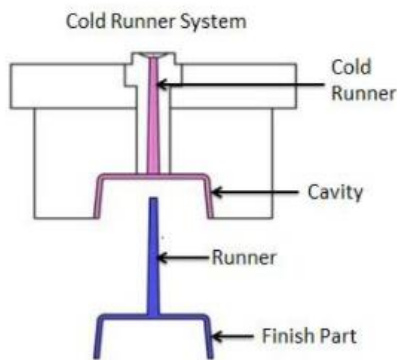
- **Hot runner**

Hot runner injection molds consist of a heated physical channel to direct molten plastic into the injection mold cavity. In this system, the runners are part of the injection mold. Temperature is kept above the material melting point.



- **Cold runner**

Cold runner injection molds consist of an unheated physical channel to direct molten plastic into the injection mold cavity after it leaves the nozzle. In this system, the runner is cooled and ejected every cycle with the part. After the complete injection cycle runner is removed from the molded part.



- **Number of Cavities**

- **Single Cavity Mold**

Single cavity mold produces only one part per cycle. The injection molding tool cost for a single cavity mold is low, but the unit part cost will be high. Single cavity molds are suitable for low production volumes.



- **Multiple Cavity Mold**

Multiple cavity mold produces more than one of the same part per cycle. Cavitation is the term used to describe the number of cavities an injection mold has. There are multiple factors to consider when determining how many cavities a mold should have. A mold with more cavities is more expensive than a mold with fewer cavities, due to its larger size and more complex hot runner system, if existing, but usually the part price is lower, because in one single cycle more parts are produced. Of course, this is not always the case, because depending on the size of the parts and the number of cavities, the mold can be very large, demanding a larger press with more clamping force and injection volume and it can make the process very expensive. Also, it is important to consider the demand of parts, making more sense to have more cavities for higher demand products. There is also an inherent risk of having one single large mold with many cavities, which is in case of damage to the mold or unplanned maintenance, the mold will be off production and it can disrupt the supply chain. So, there are multiple aspects when defining cavitation of injection molds and decision must be taken on a case-by-case basis.



- **Family Mold**

Family mold produces two or more parts per injection cycle. In this type of mold, different parts are molded in the same tool, usually parts that are used in the same product. The costs are lower than single mold cavity because with one mold more than one part can be molded. For example, if a product requires a top cover and a bottom cover and they are made of the same material with similar sizes and masses, family mold is an option to reduce costs.

Although family molds look like a good option to minimize initial investments, it is important to emphasize that there are some disadvantages as well. For example, if one of the parts is not usable, there will be an excess of the other part. There are also restrictions on the size and volume of the parts, which must be very similar. Any unbalance on weight and thickness will affect injection and cooling time, which may cause parts to warp or other defects. A Mold Flow analysis is highly recommended before proceeding with family tools.



2.3. TYPES OF GATES

A critical component in mold design is choosing the correct gate. The function of the gate is a simple concept, but choosing the wrong type of gating can cause several problems during processing. This section will describe the several different kinds of gate designs and how they are used.

There are important factors to consider when choosing the gate type/location for a molded part. The first is how the mold will be designed and where a gate can be located. Part orientation in the mold, show surface location, and action locations can limit the options for gating. Further design considerations are material selection, part volume/size and production functionality.

Some plastics are more sensitive to over-heating conditions. This is known as shear heating. Any time plastic is forced through a restricted area, there is some amount of shear heating. Attempting to fill a large plastic part with a small gate will create excessive shear heating and could degrade the plastic. For example, gate designs like submarine gates have limitations to the maximum size of the gate, which may eliminate the gate design as a possibility for larger parts.

Gate freeze is an important aspect of the injection molding process. Generally speaking, gates need to be large enough to fill out the cavity properly but small enough to seal the cavity (gate freeze). Gate

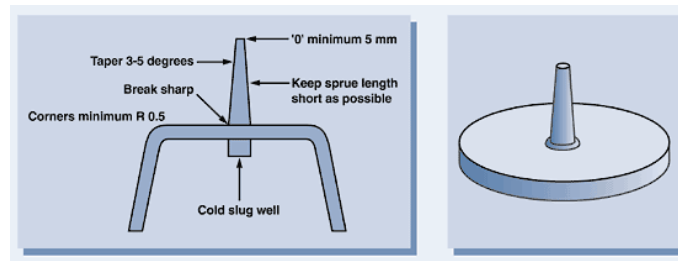
freeze should occur when the cavity is completely packed out and ready to start the cooling process. Since the gate is small, it should cool off relatively quickly compared to the molded part or the feed system. Gate freeze allows the part to become isolated from the feed system and for the injection molding machine to move onto the cooling/charging phase of the molding process.

The function of the mold in the production environment is another consideration. A defining characteristic of the different kinds of gates is whether they break from the molded part automatically or require trimming. In higher volume applications, it becomes less feasible for an operator to trim gates manually. In this case, either factoring in robotic gate cutting (Nipping) or changing gate designs should be factored in. Scrap costs generated from cold runner systems may justify a hot runner where zero scrap is generated.

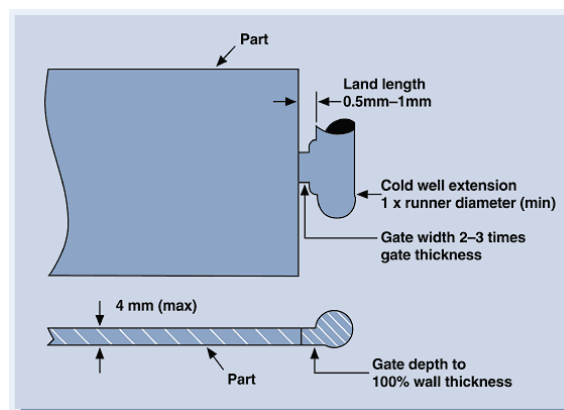
The most common gates types in injection molding are:

- **Manually trimmed**

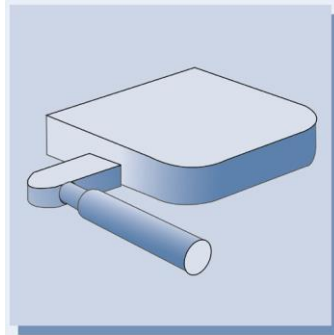
- **Sprue gate** - this gate is suitable for single chamber parts and thick sections because it can hold pressure well. The disadvantage of this gate is that a mark is left on top.



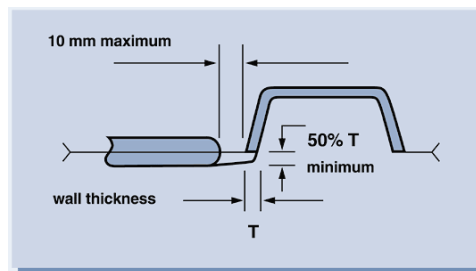
- **Edge or side gate** - this is commonly used for molds that have two or more cavities. It is placed at the side of the molds and the gate is done manually by a cutter



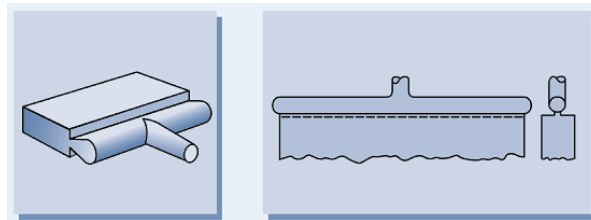
- **Tab gate** - this is another common gate style that contains an auxiliary tab section where shear stress can safely be absorbed without affecting the quality of the part. It is suitable for thin and flat parts



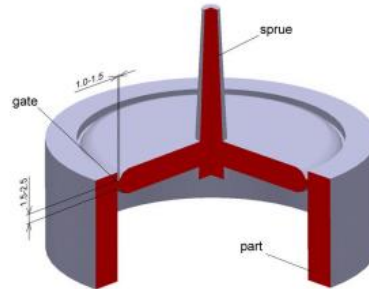
- **Overlap gate** - this gate is similar to an edge gate, except the gate overlaps the wall or surfaces. This type of gate is typically used to eliminate jetting.



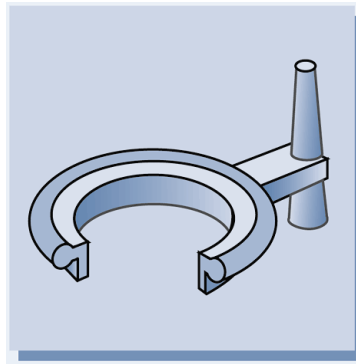
- **Film or flash gate** - this gate consists of a straight runner and a gate land across either the entire length or a portion of the cavity. It is used for long flat thin walled parts and provides even filling.



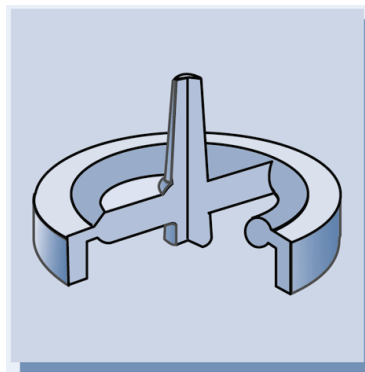
- **Diaphragm gate** - this gate is often used for gating cylindrical or round parts that have an open inside diameter. It is used for single cavity molds that have a small to medium internal diameter. It is used when concentricity is important and the presence of a weld line is not acceptable or when an even plastic flow is desired. Having even plastic flow helps to ensure that, when cooling, the part shrinks as consistently as possible and reduces stress on the molded part.



- **External ring** - this gate is used for cylindrical or round parts in a multiple cavity mold or when a diaphragm gate is not practical. Material enters the external ring from one side forming a weld line on the opposite side of the runner this weld line is not typically transferred to the part.

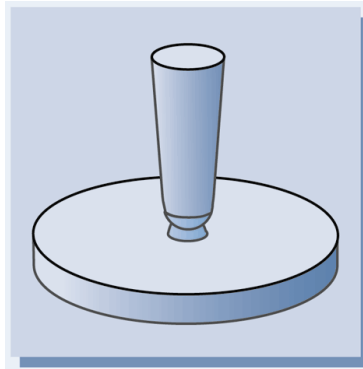


- **Internal ring gate** - this gate is used for cylindrical or round parts in a multiple cavity mold or when a diaphragm gate is not practical. Material enters the ring from inside forming a weld line on the opposite side of the runner this weld line is not typically transferred to the part.

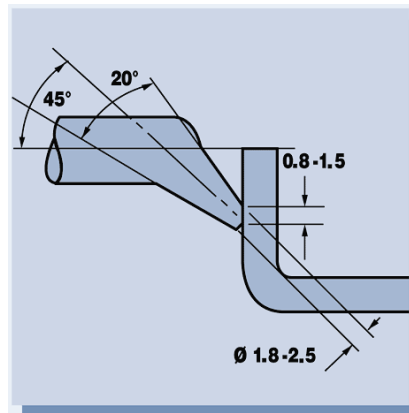


- **Automatically trimmed**

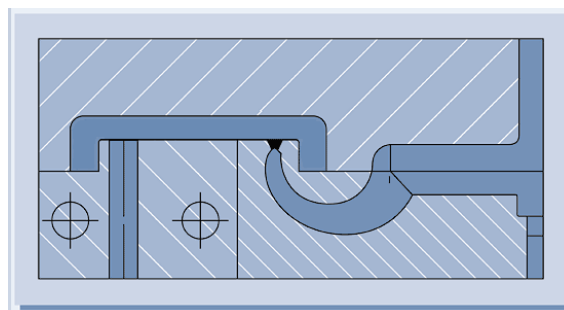
- **Pin gate** - this gate is only feasible with a 3-plate tool because it must be ejected separately from the part in the opposite direction. The gate must be weak enough to break off without damaging the part. This type of gate is most suitable for use with thin sections.



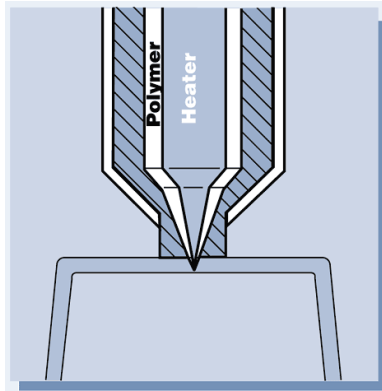
- **Submarine (tunnel) gate** - this gate is used in two-plate mold construction. An angled, tapered tunnel is machined from the end of the runner to the cavity, just below the parting line. As the parts and runners are ejected, the gate is sheared at the part. The tunnel can be located either in the moving half or in the fixed half. A sub-gate is often located into the side of an ejector pin on the non-visible side of the part when appearance is important.



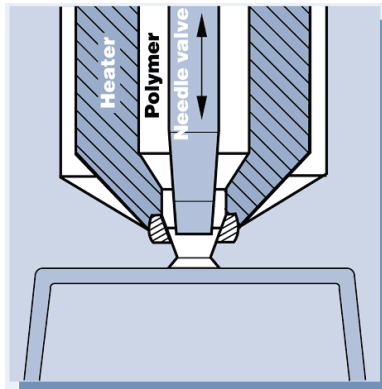
- **Cashew gate** - this is a variation of the tunnel gate design and consists of a curved tunnel gate where the tunnel is machined in the movable mold half.



- **Hot runner gate** - this gate is also known as sprueless gate. It is used to deliver hot material through hot runners into the cavity. The nozzle of a runnerless mold is extended forward to the part and the material is injected through a pinpoint gate. The face of the nozzle is part of the cavity surface; this can cause appearance issues (matte appearance and rippled surface). The nozzle diameter should be as small as possible. Most suitable for thin walled parts with short cycle times, this avoid freezing of the nozzle.



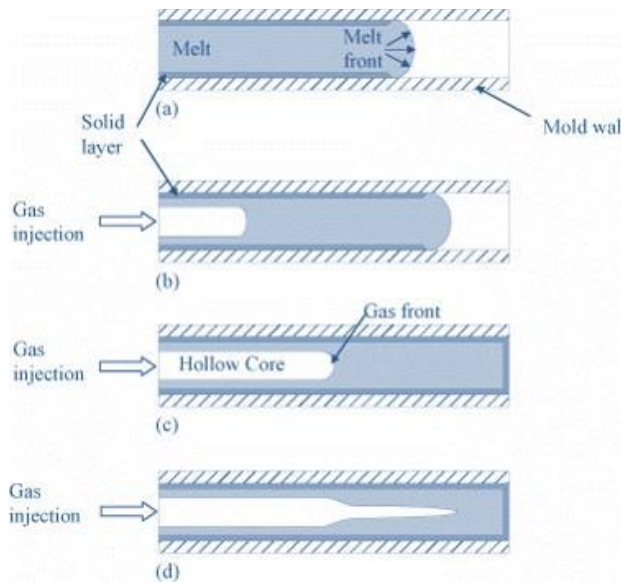
- **Valve gate** - this gate adds a valve rod to the hot runner gate. The valve can be activated to close the gate just before the material near the gate freezes. This allows a larger gate diameter and smooths the gate scar. Since the valve rod controls the packing cycle, better control of the packing cycle is maintained with more consistent quality.



2.4. TYPES OF INJECTION MOLDING PROCESS

2.4.1. Gas Assisted Injection Molding

Gas assisted injection molding consists of a process that introduces nitrogen gas into a mold cavity after it has been partially filled with plastic up to 70-80% of the mold volume. The compressed nitrogen displaces a portion of the molten plastic when injected into the cavity. The result is hollow parts that are light and relatively inexpensive to make. Designers can use gas-assist molding to create thin-walled parts. Such parts can be molded with low clamp tonnage, which reduces both tooling cost and required injection molding machine size.



2.4.2. Over Molding & Insert Molding

Over molding is an injection molding process used to mold one plastic (commonly a rubber-like plastic such as TPE) over top of another part (substrate). The substrate is usually an injection molded plastic part, but various other materials could also be used. Sometimes this process is also called insert molding, especially when the substrate is a metallic part.

Over molding process can be used to combine two parts of different materials in one single part, eliminating one step of the assembly process. It can provide two-tone aesthetics to parts, or add molded sealing or gaskets to the final part, or add a grip area to handles, for example.

In the case of insert molding, when the substrate is a metallic part, the insert is assembled to the mold prior to the injection process and the molten material is molded on top of it.



2.4.3. Film Insert Molding

Film Insert Molding consists of a process that molds the plastic material on top of a plastic film, usually decorated by printing on the back. This film can be flat or pre-shaped to the required design, depending on the application.

This process is also known as In-Mold Decorating (IMD) or In-Mold Label (IML), depending on some characteristics.



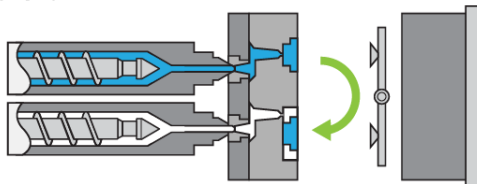
2.4.4. Double Injection Molding

Double Injection, sometimes known as 2 material 2 shot molding, is a manufacturing process used to produce complicated molded parts from two different materials. Through a highly specialized and automated process, two different materials, including two different resins (since compatible) can be molded into a single, multi-chambered mold.

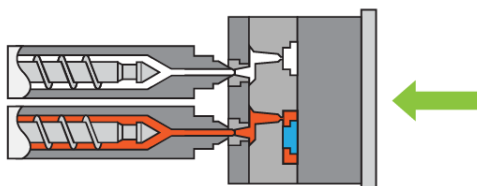
Double Injection is performed on one machine that is programmed to perform two injection shots in one cycle. In the first shot, a nozzle injects plastic into a mold. The mold is then automatically rotated and a different type of plastic is injected into the mold from a second nozzle.

Double Injection Molding leverages the compatibility between materials to create molecular bond. The result is a single part with production and feature advantages. It can be used for a variety of product designs across all industries. It also allows for molding using clear plastics, colored graphics and stylish finishes which improves your product functionality and marketplace value.

1 First Shot



2 Second Shot



3. **SURFACE FINISHES**

After rough machining there are always tool marks left on the mold surface. If these marks remain untreated, they would get transferred to the part during molding. Surface finishing requirements usually depend on the part application, but even within the same part, there might be different requirements for different locations or sides. For instance, parts or portions of the parts may be non-visible or non-aesthetic, other parts or portions of the parts may require high gloss or a specific texture. So different surface finishing treatments are necessary to meet those requirements.

It is also important to mention that depending on the machining method, more or less polishing or finishing will be necessary. For example, rough machining leaves much more marks than finish machining. If wire EDM (Electrical Discharge Machining) milling is used, it is very likely that polishing or finishing will not be needed because of the nature of this process that produces a very smooth surface on the tool.

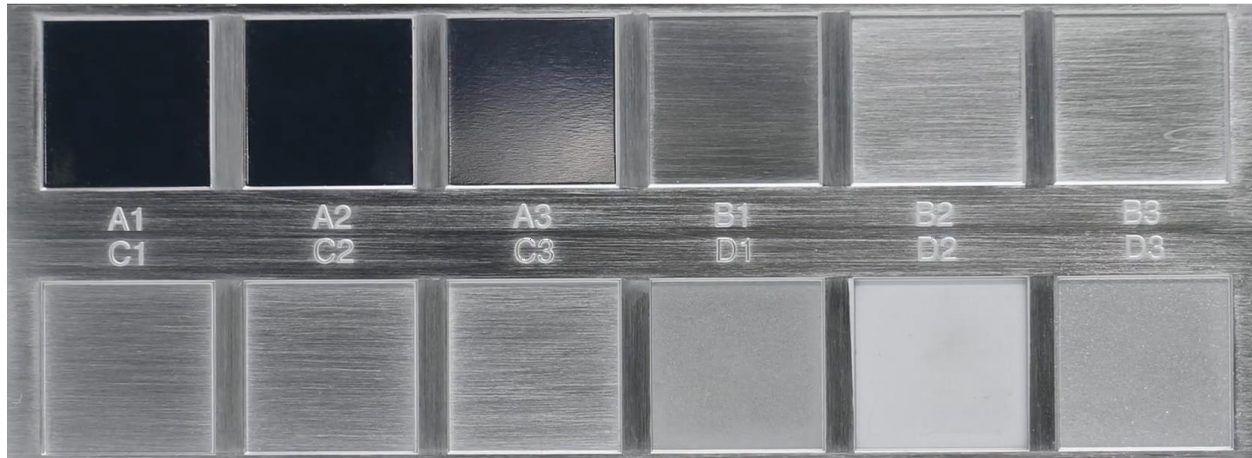
Surface finishes, such as textures, are also an effective way of hiding molding defects. For example, if there is an aesthetic area that contains a wall on the other surface and therefore is susceptible to sinking, a texture can be added so it will help hiding potential sink marks.

There are several surface finish options for injection molded plastic parts. Surface finishing procedures for injection molding can help to either increase or decrease the roughness of a part. Plastic glossy texture may be preferred for aesthetic parts, while rougher finishes may be best for certain mechanical parts or to reduce overall costs. It is also possible to create special textures, depending on the application.

3.1. **SANDING AND POLISHING**

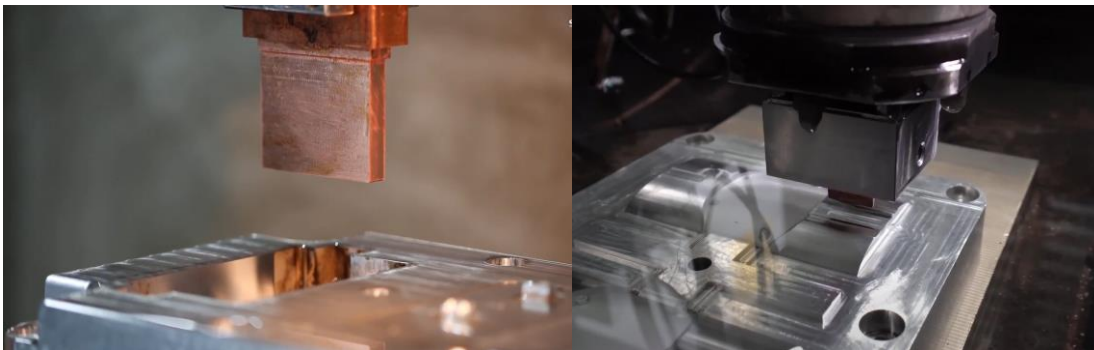
There are twelve grades of finishes specified by the Plastics Industry Association (PLASTICS, former SPI) in four categories that range from shiny to dull. Each grade has different requirements for allowable deviation from perfect, with lower numbers allowing for less deviation and higher numbers allowing for more deviation.

Finishing Standard	Surface Roughness Ra (µm)	Finishing Method	Finishing Look
A-1	0.012 to 0.025	6000 Grit Diamond	Super High Glossy finish
A-2	0.012 to 0.025	3000 Grit Diamond	High Glossy finish
A-3	0.05 to 0.10	1200 Grit Diamond	Normal Glossy finish
B-1	0.05 to 0.10	600 Grit Paper	Fine Semi-glossy finish
B-2	0.10 to 0.15	400 Grit Paper	Medium Semi-glossy finish
B-3	0.28 to 0.32	320 Grit Paper	Normal Semi-glossy finish
C-1	0.35 to 0.40	600 Grit Stone	Fine Matte finish
C-2	0.45 to 0.55	400 Grit Stone	Medium Matte finish
C-3	0.63 to 0.70	320 Grit Stone	Normal Matte finish
D-1	0.80 to 1.00	Dry Blast Glass Bead	Satin Textured finish
D-2	1.00 to 2.80	Dry Blast #240 Oxide	Dull Textured finish
D-3	3.20 to 18.0	Dry Blast #24 Oxide	Rough Textured finish



3.2. EDM SPARK EROSION

Electrical Discharge Machining erosion uses a graphite or copper electrode placed in an electrolytic bath of water or oil. The electrode discharges a spark that strikes the tool wall. This process is ideal for making deep thin slots, sharp concave corner and other features. A very fine and smooth finish can be achieved, which would be time-consuming and expensive if done manually.



3.3. MEDIA BLASTING

In this process, high pressure air is used to spray various types of dry or wet abrasive media against the tool wall. Examples of these media would be silica (sand), aluminum oxide, glass or plastic beads. This method is commonly used to clean the tool and at the same time create a uniform matte or satin finish.

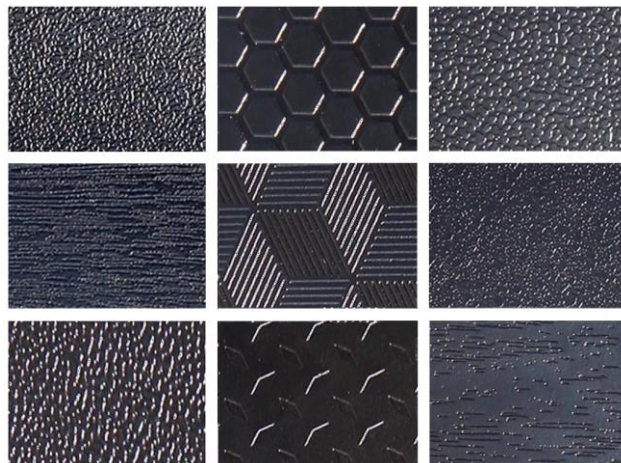




3.4. CHEMICAL PHOTOETCHING

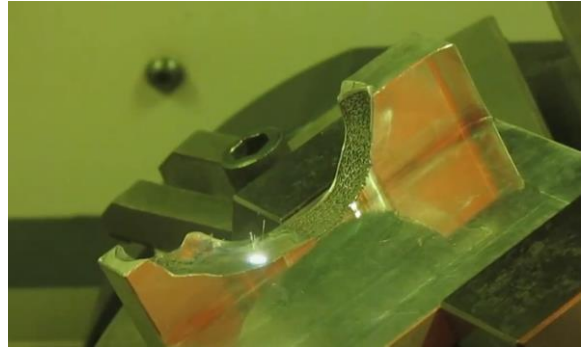
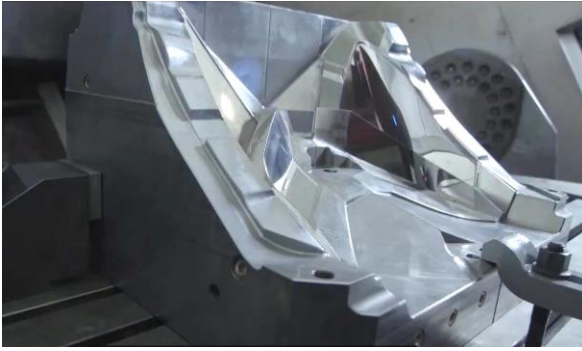
In this process, the tool wall is coated with a photoresist or light sensitive chemical. The desired pattern is optically projected onto the tool surface and any areas struck by UV light are cured. The remaining photoresist get washed away, leaving behind a film mask. The mold is then bathed in acid, which etches away the unprotected areas and creates the desired texture.

This is a great method to create sophisticated in-molded patterns and textures. It can imitate the look of leather, stone, wood grain or even abstract geometric designs.



3.5. LASER ETCHING

This process allows to create almost any texture on any surface due to its flexibility. Usually robotic arm with 6 axes of movement is used to apply the laser accurately on the mold surface. It can also reach undercuts and other hidden areas. This method is most often used in large parts, such as automotive dashboards, which have fine patterns that are consistent over a long length.



4. PART DESING GUIDELINES

This section describes important design characteristics that need to be considered when designing a plastic part for injection molding. It also contains a table with basic specifications for plastics tolerances.

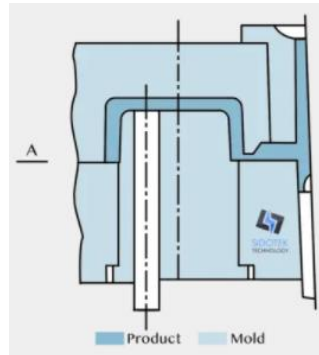
4.1. PARTING LINE

Parting line is an inevitable part of injection molded products. It is the line of separation between where two halves of a mold meet and it runs the entire perimeter of the part. It is also where a slide mechanism might meet where it shuts off on one of the mold halves. The parting line will always be visible, but can be minimized by its location, draft allowance and part geometry.

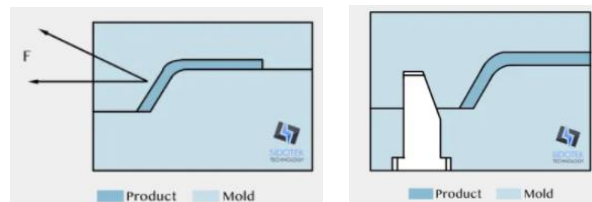


There are 5 types of parting lines:

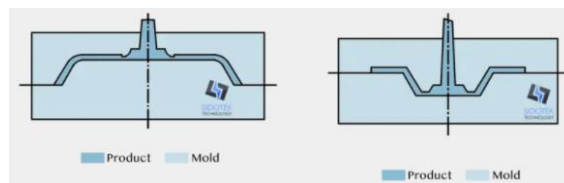
- **Vertical parting lines** - this is the most common type of parting line. It is a plane perpendicular to the mold opening direction.



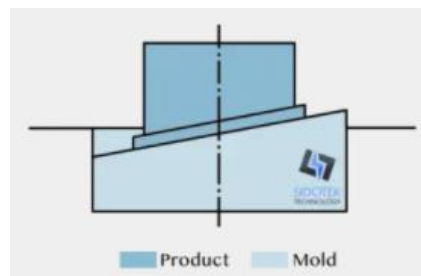
- **Stepped parting line** - depending on the part geometry, the parting line can be stepped. In this case, an unbalanced force is generated on both sides of the cavity, which can cause the halves of the mold to slide. This can be overcome using pins to support the force.



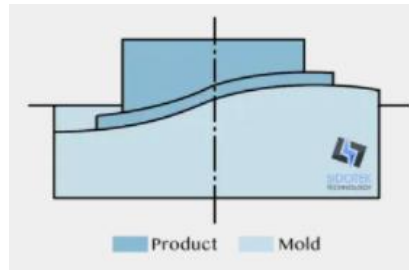
It is also possible to eliminate the unbalancing with a symmetrical arrangement of the cavities.



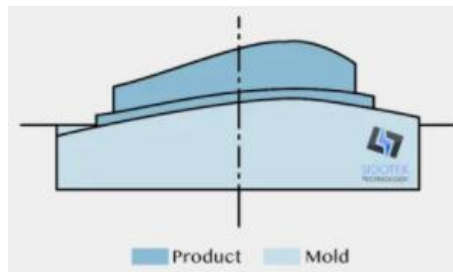
- **Beveled parting lines** - this is similar to stepped parting line, but with a beveled shape



- **Curved parting lines** - this is similar to stepped parting line, but with a curved shape



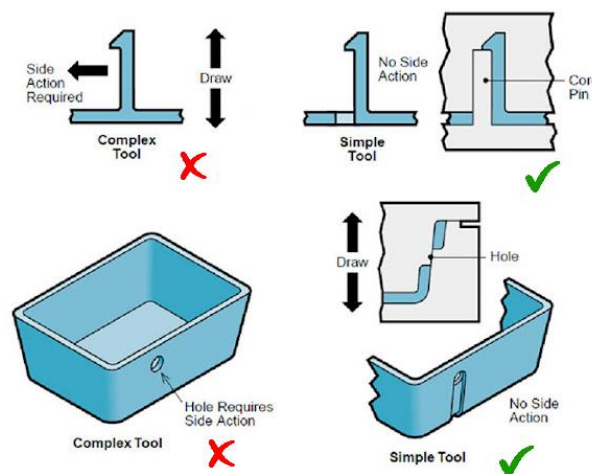
- **Comprehensive parting lines** - this form is designed based on the structure of the plastic part, it can combine all other parting line forms.



4.2. UNDERCUTS

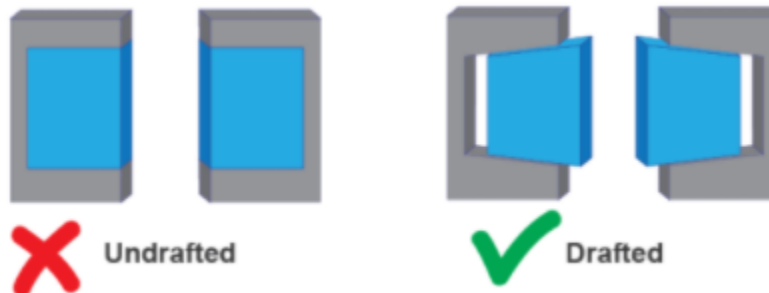
During the design phase of an injection molded part it is important to consider that the plastic goes into the mold as liquid it cools rapidly and solidifies into the shape of the cavity. Designing a part that can be ejected using a “straight pull” is key to keeping the cost of the injection mold down. A straight pull mold is designed so that when the two halves separate from each other, there is no metal blocking the path of the plastic in the direction of the pull. If the part requires a snap-fit, a hole or any detail on the sides of the part, mold will require slides and lifter mechanisms to allow the part to be ejected.

Avoiding undercuts might be the best option. Undercuts always add cost, complexity, and maintenance requirements to the mold. A clever redesign can often eliminate undercuts.



4.3. DRAFT ANGLE

A draft angle is the amount of taper that is applied to each side of most features of an injection molded part. The angle, which is positioned to run toward the direction of a mold’s pull and parting line, helps with releasing the part from the mold. Most of the times draft is required to allow a successful ejection of the molded part. There are a very few cases where draft may not be needed, such as low-friction materials or special features like crush ribs.



Having draft is crucial to manufacturing parts that meet quality standards. Without draft, major problems can arise in the injection molding process that increase the time and cost of production.

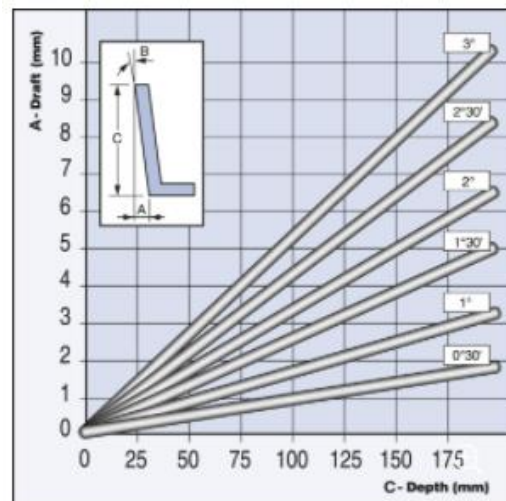
When no draft is designed into the part, problems can come from friction and vacuums. These issues can damage both the part and even the mold in extreme cases. These problems are a direct result of having insufficient draft. Because plastic shrinks as it cools, portions of the part may pull away from the mold while others grip the core. If there is no draft angle that allows a simple push from the ejector pins to pop the part out of the mold, the part's surfaces will often drag along the mold as it is being ejected. This friction between the two surfaces can cause scratches or gouges in the part and affect the appearance. This is particularly troublesome if your part is being molded with a surface finish or texture. The friction will scrape away texture and leave you with many rejected parts.

The recommended draft angle depends on many factors:

- **Material shrinkage** - some materials shrink more than others. In general, the higher the shrinkage, the larger the draft angle needed to prevent ejection issues.
- **Height and shape of wall or feature** - a shallow single straight rib having not enough draft causes less problems than a high cylindrical wall.
- **Surface texture** - a polished surface requires a smaller draft angle than a surface that is textured.
- **Aesthetic requirements** - scuff marks might not be as big of a problem on a technical part "hidden" in an assembly while, for the cover part of the assembly, they may be totally unacceptable.

The figure below shows a guideline for draft angles, considering all the above factors:

- Typically, one to three degrees draft is recommended.
- For untextured surfaces, generally a minimum of 0.5 degree is recommended.
- For textured surfaces on sidewall, an additional 0.4-0.5 degree is recommended per 0.1mm of texture depth.



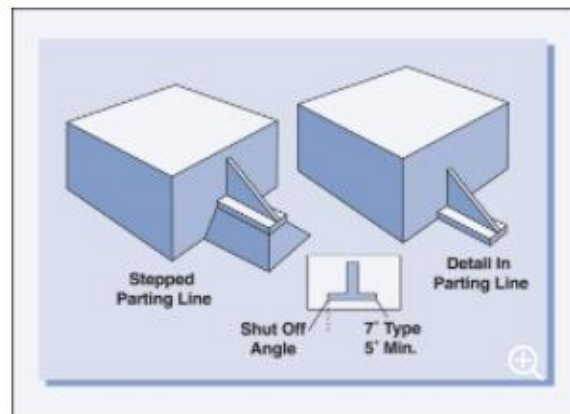
Draft in mm for various draft angles as function of depth

4.3.1. Draft Angle for Shut-offs

The most straightforward mold construction is where the parting plane of the mold halves is a flat surface perpendicular to the draw direction. This setup also offers the easiest way to make sure that there is no gap between the mold halves when they are closed. After all, the clamping force of the injection molding machine acts perpendicular to the parting plane. However, it is not always possible or even beneficial to keep the parting plane flat.

In case of such a stepped parting line, the parting line jumps from one level to another. The surface needed to bridge these levels is called a shut-off because it is where the mold halves shut against each other. The shut-off should never be exactly parallel to the draw direction. This leads to drag, causing the tool to wear over time. In a worst case scenario, misalignment during mold closing can cause significant damage. Instead, draft should be applied to guarantee proper shut-off.

On stepped parting surfaces, a draft of seven or more degrees is always recommended. Five degrees should be considered as a minimum.



4.4. RIBS, CORRUGATIONS AND GUSSETS

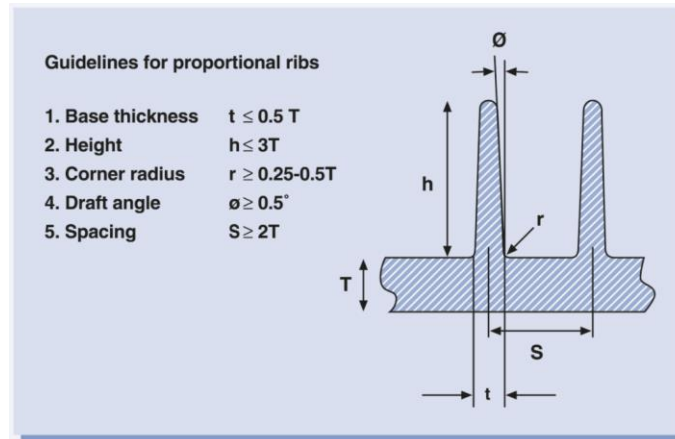
When the stiffness of a plastic part needs to be improved, it is necessary to either increase its sectional properties or change the material. Sometimes, it may be enough to change the material grade, for example using a higher glass fiber content. If that is not sufficient or desired for other reasons, increasing the sectional properties is often the solution.

In many cases, the simplest way to improve the sectional properties is increasing wall thickness. However, this has some limitations. Additional reinforcement can be obtained by adding corrugations or placing ribs perpendicular to the parts wall. Adding corrugations typically has a smaller impact on part weight and cooling time, but adding ribs has a bigger potential in terms of increasing the stiffness. Ribs allow for the facing or mating surface of a part to be smooth, which can be beneficial for aesthetic or functional reasons.

4.4.1. Ribs

Rib dimensioning

When adding ribs, the following guidelines for dimensioning should be applied:



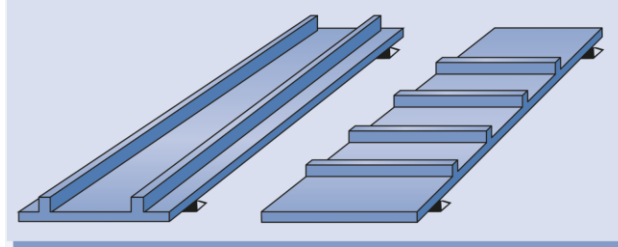
- The thickness of the rib should be approximately 50 to 60% of the general wall thickness of the part. Exceeding this value may lead to sink marks on the surface opposite the ribs. It can negatively influence material flow during injection, possibly resulting in weld lines and voids.
- The rib height should not exceed three times the general wall thickness as deep ribs become difficult to fill and may stick in the mold during ejection.
- On the sides of ribs, a draft angle of 1 - 1.5 degree should be applied. This means that the ribs are slightly tapered, becoming slightly thinner towards the top. This makes ejection of the part from the mold easier. On low ribs and in exceptional cases, a smaller or even no draft angle is acceptable. It should be noted though, that this may lead to cosmetic defects like scuff marks or issues with ejecting the part after molding.
- At the base of the rib, where it intersects with the nominal wall, a radius of 25 - 50% of the general wall thickness should be included. A minimum radius of 0.4 mm is suggested. This will eliminate a potential stress concentration and improve flow and cooling characteristics around the rib. When exceeding the 50% value, a material mass develops, increasing the risk of molded-in residual stresses, voids or sink marks.
- Spacing between two parallel ribs should be at least two times the general wall thickness. This prevents the mold from developing a hot blade that is fragile and has cooling problems.

Rib placement

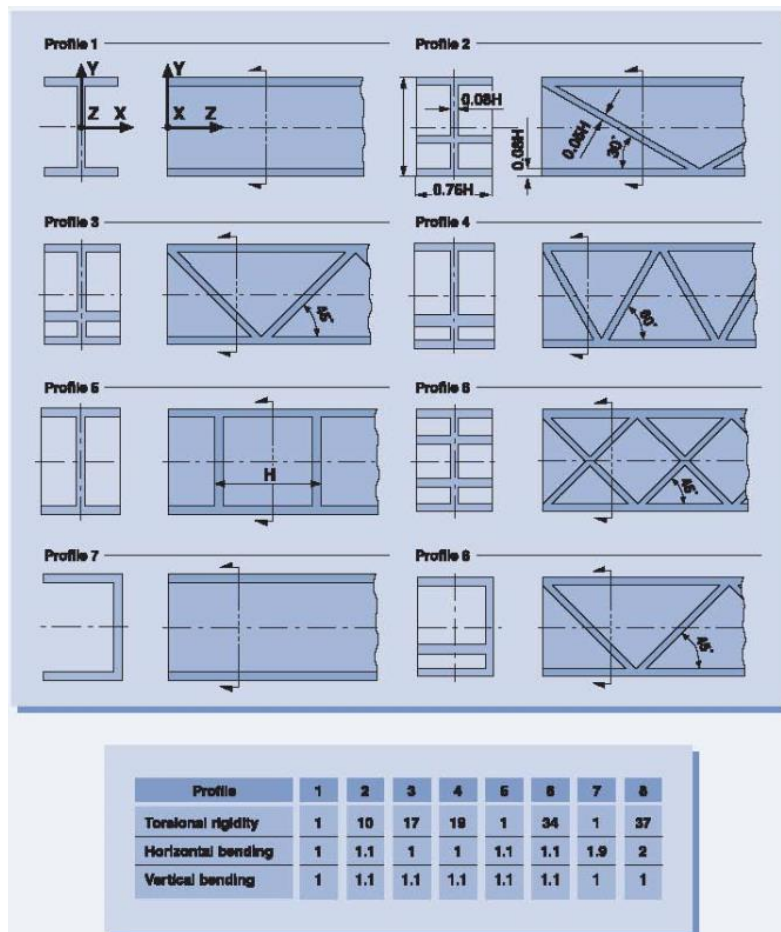
When placing ribs, the following guidelines for placement should be applied:

- Ribs are preferably designed parallel to the melt flow as flow across ribs can result in a branched flow leading to trapped gas or hesitation. Hesitation can increase internal stresses and short shots.
- Parallel ribs should be spaced at a minimum distance of twice the nominal wall thickness; this helps prevent cooling problems and the use of thin blades in the mold construction.

- Ribs should be orientated along the axis of bending in order to provide maximum stiffness. Consider the example below, where a long thin plate is simply supported at the ends. If ribs are added in the length direction, the plate is significantly stiffened. However, if ribs are added across the width of the plate, little improvement is found.



- When adding ribs to a profile or boxed section, their placement and orientation has a large impact on the obtained additional stiffness.



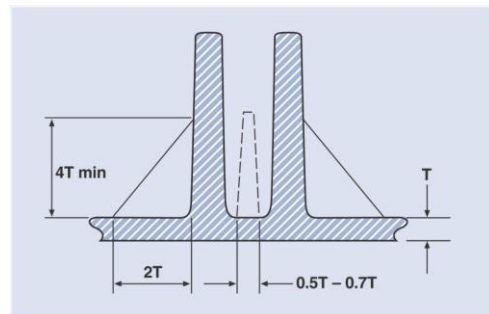
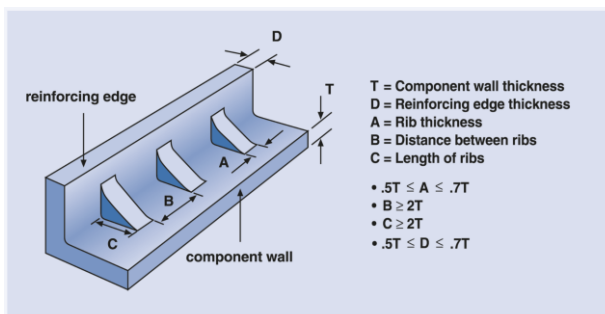
- For maximum performance and function, the neutral lines of the ribs and profile wall should meet at the same point. However, depending on the specific dimensions and material of choice, sink marks may occur. This can be avoided but will result in a weaker geometry:
 - If the diagonal ribs are moved slightly apart, the rigidity is reduced 35%.
 - If a short vertical rib is added to the design, the torsional rigidity is reduced an additional 5%

Variants	Rigidity in %	Resistance to stress in %
	100	100
	65	80
	60	82

Ratio $S_r/S_p = 0.69$

4.4.2. Gussets

Gussets can be used to reinforce corners, side walls and bosses. They can be considered as a subset of ribs, meaning that the guidelines for rib dimensioning and placement are also valid for gussets.

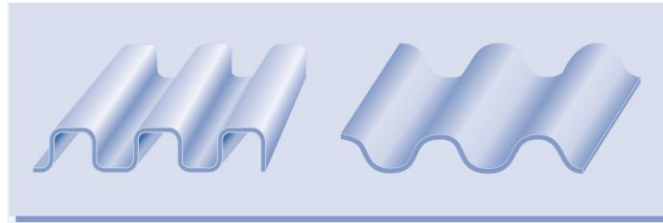


4.4.3. Corrugations

Adding corrugations to the design can stiffen flat surfaces in the direction of the corrugations. They are very efficient and do not add large amounts of extra material or lengthen the cooling time. The extra stiffness is a result of increasing the average distance of the material from the neutral axis of the part.

In order to increase stiffness in the direction perpendicular to the corrugations and improve straightness, walls can be added and the corrugation profile can be offset by about half the corrugation

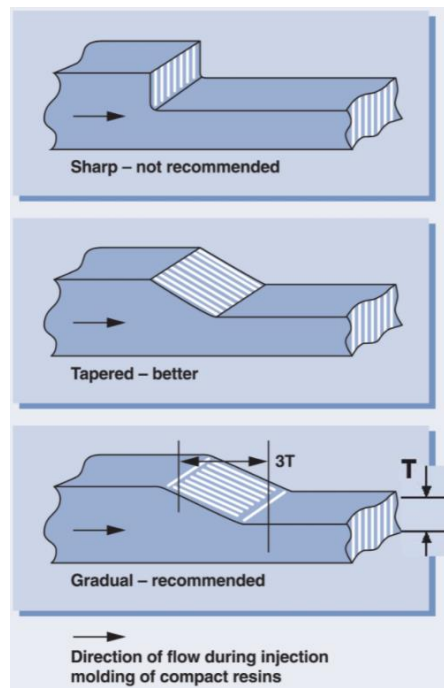
pitch. The distance between the wall will depend on the material and thickness of the part and should be determined case by case.



4.5. WALL THICKNESS

The recommended general wall thickness of injection molded parts depends on the size of the geometry, the material of choice and the desired performance of the geometry.

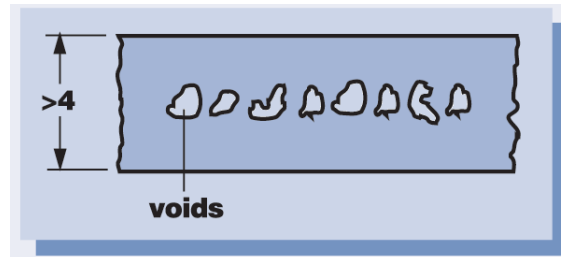
Typically, the recommended wall thickness is in the range 0.5 mm to 4 mm. In specific cases, wall thicknesses can be either smaller or bigger. A basic design guideline is to keep wall thicknesses as thin and as uniform as possible. Where varying wall thicknesses are unavoidable for reasons of design, there should be a gradual transition as indicated below:



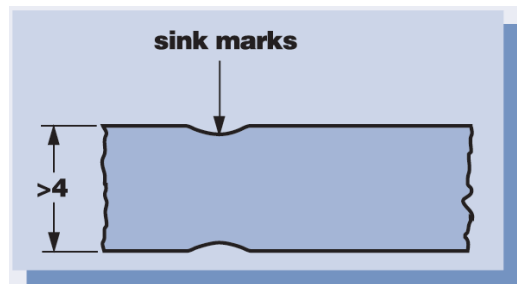
The wall thickness is important because of its influence on:

- **Mold filling** - if the wall thickness doesn't fit the flow behavior of the thermoplastic material, it may be hard to completely fill the mold.
- **Part weight** - the greater the wall thickness, the heavier the part.
- **Cooling time** - the greater the wall thickness, the longer it will take for the part to cool down after injection molding.

- **Part cost** - both of the above, plus larger part volume and increased injection molding cycle time results in higher part cost.
- **Dimensional accuracy** - different areas of the part having different cooling rates, which is typically the case when the wall thickness is high or not uniform, leads to molded-in residual stresses that cause the part to warp after being ejected from the mold.
- **Part performance** - thick sections can cause voids to arise within the wall thickness.



- **Part aesthetics** - If the wall thickness is too high, non-uniform cooling rates may lead to sink marks.



4.6. RADII AND CHAMFERS

Sharp edges and corners should always be avoided. This is because they lead to the following results:

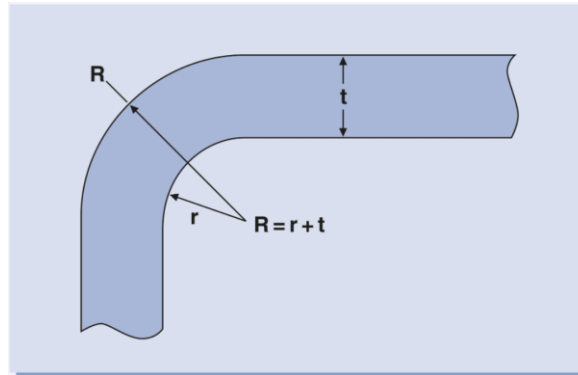
- High molded-in stresses
- Poor flow characteristics
- Reduced mechanical properties
- Increased tool wear
- Surface appearance problems

Instead, by applying radii and fillets, the following results can be achieved:

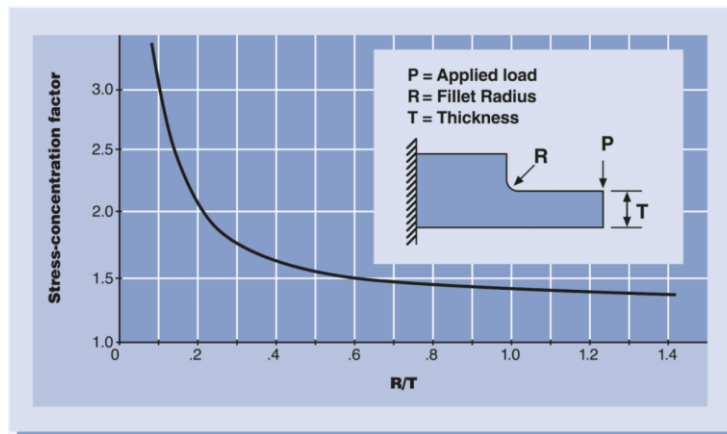
- Less warpage
- Less flow resistance and easier filling
- Reduced stress concentration
- Less notch sensitivity
- More uniform cooling
- Better appearance

For radii and fillets, these are the recommended guidelines:

- Sharp corners should be rounded with a radius measuring between 25% and 60% of the nominal wall thickness, depending on the case.
- A minimum inner corner radius of 0.5 mm is suggested.
- In order to keep the wall thickness uniform, the outside radius of a corner should be equal to the inner radius plus the wall thickness.



- All sharp outer part and rib edges should be broken with at least a 0.125 mm radius.
- For a part with an inner corner radius half the nominal wall thickness, a stress concentration factor of 1.5 is a reasonable assumption. For smaller radii, for example 10% of the nominal wall, this factor will increase to 3. Standard tables for stress concentration factors are available and should be consulted for critical applications.



Like a radius or fillet, a chamfer can be used to soften sharp edges or make a gradual transition between two perpendicular faces. In view of minimizing stress concentration and optimizing flow, a radius is better than a chamfer. However, a chamfer can offer additional functionality like facilitating manual positioning of parts during assembly. A chamfer can also be used from an aesthetics point of view: where many parts are rounded-off, a chamfered edge creates a distinctive look. As with radii, a uniform wall thickness is preferred, so a chamfer on an outer edge should be copied on the inner corner.

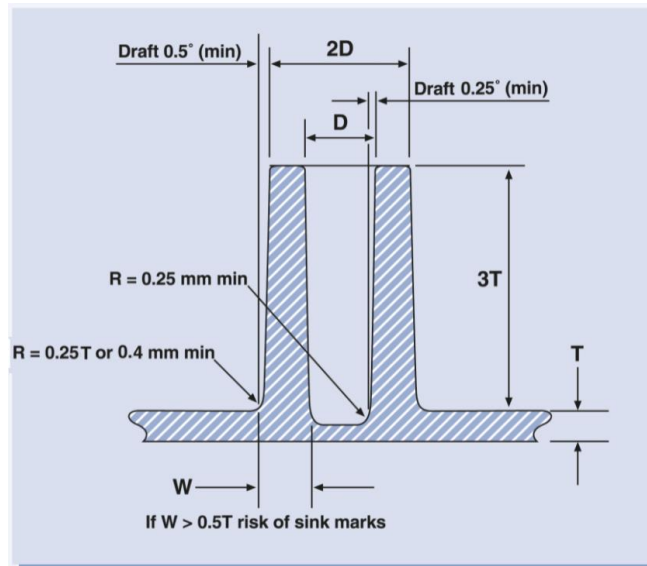
4.7. BOSSES

A boss is a cylindrical protrusion placed on a part's wall. A boss can have various functions, such as the following:

- Positioning aid - To help align parts during assembly; for example, a pin on one part will fit a hole in the other part.
- Fixation point - In the example mentioned above, the pin and hole can have a press fit. Friction between the two keeps the parts fixed. A boss can also be a tubular protrusion that can accommodate self-tapping screws to fix other parts.
- Bearing surface - A cylindrical protrusion can act as a bearing for another part; for example, a lever or a gear to rotate about.

The dimensions of a boss depend on its function. Manufacturability and aesthetics should be considered, for example, thick sections need to be avoided as these can lead to an increased injection molding cycle time, non-uniform shrinkage, sink marks and molded-in stresses. If the boss is to be used to accommodate self-tapping screws or inserts, the wall section must be controlled to avoid excessive stress in the boss.

General recommendations regarding boss dimensions include the following:

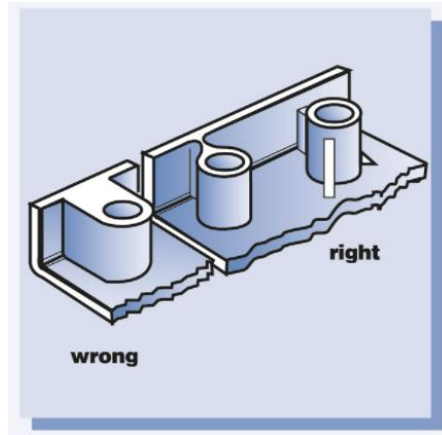


- The wall thickness of a boss should be less than 50% of the general wall thickness.
- A radius of 25% of the general wall thickness or 0.4 mm should be applied at the base of the boss to avoid stress concentration.
- A minimum draft of 0.5 degrees is required on the outer surface of the boss to facilitate release from the mold on ejection.
- In case of a tubular boss, the risk of sink marks can be reduced by extending the core pin so that it slightly penetrates the wall the boss is placed on. A minimum radius of 0.25 mm should be applied to the edge of the core pin to reduce material turbulence during filling and to help keep stresses to a minimum.

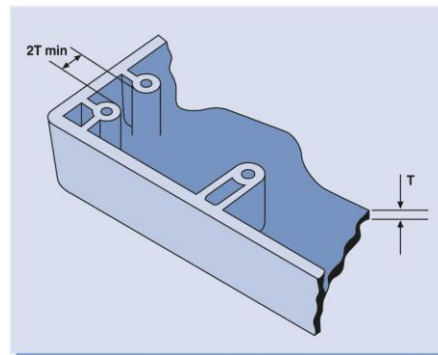
- A minimum draft of 0.25 degrees is required on the inner surface of a tubular boss for ejection and or proper engagement with a fastener.

The strength of a boss can be increased by adding gussets or by connecting it to a sidewall by adding a rib. In that case:

- Bosses adjacent to external walls should be positioned a minimum of 3 mm from the surface to avoid local material accumulation.



- Regarding spacing between bosses, a minimum distance of twice the nominal wall thickness should be used.



Note: For cases like this, where steel features will be exposed to bending forces, it is recommended a maximum aspect ratio of 3:1 (height:width) to avoid stress during the fill cycle and tension cycling during the eject cycle. For higher ratios it is also recommended that these features should be aligned with the flow direction of the molten polymer.

4.8. DEFAULT SPECIFICATIONS FOR PLASTIC TOLERANCES

Feature Size	0-10 mm	10.1-25 mm	25.1-50 mm	50.1-75 mm	75.1-100 mm	100.1-150 mm
Toolbound Commercial Tol (mm)	+/- 0.100	+/- 0.125	+/- 0.150	+/- 0.200	+/- 0.225	+/- 0.250
Toolbound Fine Tol (mm)	+/- 0.050	+/- 0.075	+/- 0.100	+/- 0.125	+/- 0.150	+/- 0.175
Non-Toolbound Commercial Tol (mm)	+/- 0.200	+/- 0.225	+/- 0.250	+/- 0.300	+/- 0.350	+/- 0.400
Non-Toolbound Fine Tol (mm)	+/- 0.100	+/- 0.125	+/- 0.150	+/- 0.175	+/- 0.200	+/- 0.225

Feature Size	150.1-200 mm	200.1-250 mm	250.1-300 mm	300.1-350 mm	350.1-400 mm	400.1-450 mm
Toolbound Commercial Tol (mm)	+/- 0.500	+/- 0.650	+/- 0.800	+/- 0.950	+/- 1.100	+/- 1.250
Toolbound Fine Tol (mm)	+/- 0.275	+/- 0.375	+/- 0.475	+/- 0.575	+/- 0.675	+/- 0.775
Non-Toolbound Commercial Tol (mm)	+/- 0.600	+/- 0.750	+/- 0.900	+/- 1.050	+/- 1.200	+/- 1.350
Non-Toolbound Fine Tol (mm)	+/- 0.325	+/- 0.425	+/- 0.525	+/- 0.625	+/- 0.725	+/- 0.825

Toolbound: Dimensions that are contained in a single side of the tool, within in a single or non-moving piece of the tool

Non-Toolbound: Dimensions that are across moving sections of the tool, including: parting line, lifter, slide, cavity to core

REVISION HISTORY

Rev.	ECO #	Description of Change	Sections Affected	Originator/Department
01	DECO-03653	Initial Release	All	Gustavo Frattini / AME